

[CILab]

Impact Continues Integration Lab

Introduction: In software engineering, continuous integration (CI) is the practice of merging all developers' working copies to a shared mainline several times a day [Tho]. Grady Booch first proposed the term CI in his 1991 method although he did not advocate integrating several times a day [Boo]. Extreme programming (XP) adopted the concept of CI and did advocate integrating more than once per day – perhaps as many as tens of times per day [Bec99].

Objectives:

- Understand what CI is.
- Setup a simple CI pipeline.

Classwork:

1. Go to [Building and testing Java with Maven]
2. Add a `*.yaml` file to your repository path `<YOUR_PROJECT>/.github/workflows/` to tell GitHub Actions CI what to do.
3. Configure the `*.yaml` to automatically build your project for each pull request (use maven).
4. To use shared jars from GitHub Packages you need create a `.maven-settings.xml` file in the project root folder, see [Working with the Apache Maven registry]
5. Configure the `*.yaml` to execute tests automatically.

References

- [Bec99] Kent Beck. Embracing change with extreme programming, 1999. [HTML].
- [Boo] Grady Booch. Object Oriented Design: With Applications. [HTML].
- [Tho] ThoughtWorks. Continues Integration. [HTML].